Certified Log-Concavity of the Riemann–Jacobi Kernel and a Pólya-Type Criterion for Real Zeros

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Abstract

We certify strict log-concavity of the Riemann–Jacobi kernel $\Phi(u)$, appearing in the Fourier cosine representation $\Xi(t) = \int_0^\infty \Phi(u)\cos(tu)\,du$ of the Riemann Xi function. The implication from log-concavity to real-rootedness of Ξ is isolated as a Pólya-source dependency. A direct reading of the scanned original Pólya (1927)—the Göttingen digitisation (LOG Id: LOG_0006)—reveals that Satz I is a *preservation theorem* (requiring prior real-rootedness of the input integral) and Satz II is a *Mellin–Fourier theorem* (requiring the Mellin transform to have only real negative zeros); neither theorem contains the strong log-concavity criterion used in Theorem 1. The source audit yields **Verdict D (mismatch)**, documented in Section 2. Accordingly, **no unconditional Riemann Hypothesis claim is made**.

The unconditional contribution of this paper is **Theorem 15: certified strict log-concavity of Φ on** $[0, \infty)$, combining: (1) an algebraic core showing $(\log \varphi_1)''(u) < 0$ for all $u \geq 0$; (2) rigorous interval arithmetic certifying $Q_\Phi(u) < 0$ on $[0, 1]$ across 52,898 subintervals at 60-digit precision; (3) a no-cancellation approach certifying $(\log \Phi)''(u) < 0$ on $[1.0, 3.0]$ across 101 overlapping interval checks; and (4) an explicit monotonicity tail bound ($C = 204$) covering $[3.0, \infty)$.

Under a Pólya-type criterion matching the verified hypotheses (Theorem 1, attributed to Csordas–Varga (1989) [2] pending direct primary-source verification), the RH follows as a conditional consequence (Remark 16).

All 36 systematic tests detected no inconsistency. All results are reproducible at `https://github.com/BitConcepts/riemann-solver`.

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1 Introduction

The Riemann Hypothesis (RH), posed by Bernhard Riemann in 1859, asserts that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\mathrm{Re}(s) = 1/2$. Equivalently, all zeros of the Riemann Xi function

$$\Xi(t) := \xi\left(\tfrac{1}{2} + it\right), \quad \xi(s) := \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \tag{1}$$

are real. It is well known [12] that $\Xi(t)$ admits the Fourier cosine representation

$$\Xi(t) = \int_0^\infty \Phi(u)\cos(tu)\,du, \tag{2}$$

where the *Riemann–Jacobi kernel* is

$$\Phi(u) = 4\sum_{n=1}^{\infty} \varphi_n(u), \quad \varphi_n(u) = (2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2})e^{-\pi n^2 e^{2u}}. \tag{3}$$

We verify computationally that $\int_0^\infty \Phi(u)\,du = \xi(1/2) \approx 0.4971$ to 15 digits.

Approach. We certify a strong structural property of the Riemann–Jacobi kernel: strict log-concavity of Φ on $[0,\infty)$ (Theorem 15). Whether this property implies RH depends on the exact force of a Pólya-type theorem, audited in Section 2. We verify that Φ satisfies all of the hypotheses of Theorem 1 (a Pólya-type criterion from Csordas–Varga [2]); subject to the cited Pólya-type criterion, this yields RH as a conditional consequence. The direct reading of Pólya (1927) reveals a hypothesis mismatch (Verdict D), so no unconditional RH claim is made in this paper.

Prior work. The log-concavity of the dominant term φ_1 was observed by Csordas and Varga [2]. The Turán inequalities for the Taylor coefficients of Ξ were established by Csordas, Norfolk, and Varga [1], and higher-order generalizations by Dimitrov and Lucas [4] and Griffin, Ono, Rolen, and Zagier [5]. The connection between log-concavity and the de Bruijn–Newman constant was explored by de Bruijn [3] and Rodgers and Tao [11]. The key distinction of this work is verifying log-concavity of the *full kernel* Φ, not just the dominant term.

2 Source-Critical Audit of Pólya (1927)

The theorem bridge between log-concavity of Φ and real-rootedness of Ξ is attributed in the literature to Pólya (1927) [10]. This section presents a direct transcription of the two main theorems from the scanned original and assesses whether either supports the required implication.

Source. G. Pólya, "Über trigonometrische Integrale mit nur reellen Nullstellen," *J. reine angew. Math.* **158** (1927), pp. 6–18. Digitisation: Niedersächsische Staats- und Universitätsbibliothek Göttingen (GDZ), LOG Id: LOG_0006. Literal title translation: "On trigonometric integrals with only real zeros."

2.1 Exact German text of Satz I (p. 7)

> *I. Die Funktion $\varphi(t)$ sei analytisch für alle reellen Werte von t und von folgender Beschaffenheit: Wenn das Integral*
>
> $$\int_{-\infty}^{+\infty} F(t)\, e^{izt}\, dt$$
>
> *nur reelle Nullstellen hat, so hat (wie auch $F(t)$ sonst beschaffen sei) das Integral*
>
> $$\int_{-\infty}^{+\infty} \varphi(t)\, F(t)\, e^{izt}\, dt$$
>
> *ebenfalls nur reelle Nullstellen. Hierzu ist notwendig und hinreichend, daß $C\,e^{\gamma^2}\varphi(it)$ eine ganze Funktion vom Geschlecht 1 sei, die nur reelle Nullstellen besitzt und für reelles t reelle Werte annimmt; C bedeutet eine nichtverschwindende, γ eine reelle, nichtnegative Konstante.*

Literal English translation of Satz I.

I. The function $\varphi(t)$ shall be analytic for all real values of t and of the following nature: if the integral $\int_{-\infty}^{+\infty} F(t)\, e^{izt}\, dt$ has only real zeros, then (whatever the nature of $F(t)$ otherwise) the integral $\int_{-\infty}^{+\infty} \varphi(t)\, F(t)\, e^{izt}\, dt$ also has only real zeros. For this, the necessary and sufficient condition is that $C\, e^{\gamma^2} \varphi(it)$ be an entire function of genus 1 possessing only real zeros and taking real values for real t; C denotes a non-vanishing constant, γ a real, non-negative constant.

Mathematical normalization. In modern terms: φ preserves real-rootedness of Fourier transforms in the class defined by condition (3) of the introduction if and only if φ belongs to the Laguerre–Pólya class $\mathcal{LP}$.

Translation notes. *Analytisch* = analytic; *ganze Funktion vom Geschlecht 1* = entire function of genus 1 (Hadamard); *nichtverschwindend* = non-vanishing. The hypothesis that $\int F e^{izt} dt$ has only real zeros is *explicit*: Satz I is a **preservation theorem**, not a generation theorem. It requires the Fourier transform of the input kernel to already have only real zeros—which is precisely the conclusion sought for Ξ—and so cannot establish real-rootedness from scratch.

2.2 Exact German text of Satz II (pp. 7–8)

II. Es sei die Funktion $f(t)$ für positive Werte von t definiert, reellwertig, absolut integrabel und für genügend großes t der Ungleichung

$$|f(t)| < B\, e^{-t^{1+\beta}}$$

unterworfen (B, β sind positive Konstanten). Ferner sei $f(t)$ in einer Umgebung des Punktes $t = 0$ analytisch. Dann stellt das Integral

$$\int_0^\infty t^{z-1}\, f(t)\, dt = H(z)$$

eine meromorphe Funktion dar. Wenn die Funktion $H(z)$ nur reelle negative Nullstellen besitzt, und q eine positive ganze Zahl ist, so besitzt die durch das Integral

$$\int_{-\infty}^{+\infty} f(t^{2q})\, e^{izt}\, dt$$

dargestellte ganze Funktion nur reelle Nullstellen.

Literal English translation of Satz II.

II. Let the function $f(t)$ be defined for positive values of t, real-valued, absolutely integrable, and for sufficiently large t subject to the inequality $|f(t)| < B\, e^{-t^{1+\beta}}$ (B, β positive constants). Furthermore let $f(t)$ be analytic in a neighbourhood of the point $t = 0$. Then the integral $\int_0^\infty t^{z-1} f(t)\, dt = H(z)$ represents a meromorphic function. If the function $H(z)$ has only real negative zeros, and q is a positive integer, then the entire function represented by the integral $\int_{-\infty}^{+\infty} f(t^{2q})\, e^{izt}\, dt$ has only real zeros.

Mathematical normalization. Let $f : (0, \infty) \to \mathbb{R}$ be real-valued, absolutely integrable, $|f(t)| \le Be^{-t^{1+\beta}}$ for large t, and analytic near 0. The Mellin transform $H(z) := \int_0^\infty t^{z-1} f(t)\, dt$ is meromorphic. **If** H has only real negative zeros and $q \in \mathbb{Z}_{>0}$, **then** $\int_{-\infty}^{\infty} f(t^{2q}) e^{izt}\, dt$ has only real zeros.

Translation notes. *Reellwertig* = real-valued; *absolut integrabel* = absolutely integrable ($f \in L^1(0, \infty)$); *reelle negative Nullstellen* = real negative zeros; *positive ganze Zahl* = positive integer. The key hypothesis is "$H(z)$ hat nur reelle negative Nullstellen"—this is a condition on the **Mellin transform** $H(z) = \int_0^\infty t^{z-1} f(t)\, dt$, not a log-concavity condition on f. Log-concavity of f does not appear anywhere in Satz II. The conclusion concerns the Fourier integral of $f(t^{2q})$, not of f directly.

2.3 Comparison of interpretations

Interpretation	Theorem form	Implies RH from log-concavity of Φ?	Status
Strong log-concavity criterion	Positivity + log-concavity + analyticity + decay $\Rightarrow$ Fourier transform has only real zeros	Yes	**Not found in Pólya 1927**
Satz I: Preservation theorem	IF $\int F e^{izt} dt$ has real zeros, THEN multiplying by $\varphi \in \mathcal{LP}$ preserves real zeros	No (circular)	Confirmed from source
Satz II: Mellin–Fourier theorem	IF Mellin transform $H(z)$ has only real negative zeros THEN $\int f(t^{2q}) e^{izt} dt$ has only real zeros	No (different hypothesis)	Confirmed from source

2.4 Required source verdict

Verdict D — Mismatch.

Neither Satz I nor Satz II of Pólya (1927) [10] directly states the strong log-concavity criterion required to conclude that log-concavity of Φ implies real-rootedness of its Fourier transform.

- Satz I: *preservation theorem*—requires the input Fourier integral to already have only real zeros.
- Satz II: *Mellin–Fourier theorem*—requires the Mellin transform $H(z) = \int_0^\infty t^{z-1} f(t)\, dt$ to have only real negative zeros. Log-concavity of f does not appear.
- The strong log-concavity criterion does **not** appear in Pólya (1927).

Consequence: The RH conditional consequence (Remark 16) is not established from Pólya (1927) as cited. The **unconditional contribution** of this paper is **Theorem 15: certified strict log-concavity of Φ on** $[0, \infty)$.

3 A Pólya-Type Criterion (Source-Critical Dependency)

Theorem 1 (Pólya-type criterion; source-critical dependency)**.** *Let $K : \mathbb{R} \to \mathbb{R}$ be an even function satisfying:*

(i) $K(t) > 0$ for all t;

(ii) $K \in L^1(\mathbb{R})$*;*

(iii) $(\log K)''(t) \le 0$ *for all* $t \ge 0$*;*

(iv) $K(t) = O(e^{-|t|^{2+\delta}})$ *for some* $\delta > 0$*;*

(v) K *is real analytic on a neighborhood of the origin.*

Then the entire function $F(z) = \int_{-\infty}^{\infty} K(t)\, e^{izt}\, dt$ *has only real zeros.*

Status: This theorem is not used as an unconditional result. The source audit (Section 2, Verdict D) establishes that neither Pólya (1927) nor Csordas–Varga (1989) Theorem 2.2 contains a matching statement (see proof reference below for details).

Proof reference and source status. Direct reading of Satz I and Satz II in Pólya (1927) [10] establishes hypothesis mismatch (Section 2, Verdict D).

A direct reading of Csordas–Varga (1989) [2] (PDF available at `https://www.math.kent.edu/~varga/pub/paper_169.pdf`) reveals that their **Theorem 2.2 is itself the Mellin–Fourier theorem (Pólya Satz II)**, not the log-concavity criterion. Specifically, Csordas–Varga Theorem 2.2 reads (paraphrase):

> If the Mellin transform $H(z) = \int_0^{\infty} t^{z-1} K_1(t)\, dt$ has only real negative zeros, then $\int_{-\infty}^{\infty} K_1(t^q) e^{izt}\, dt$ has only real zeros.

This is cited as "[23, p. 7]" in their paper, where [23] is Pólya (1927). The hypothesis is the same Mellin-transform condition as Satz II; log-concavity of K_1 does not appear.

Csordas–Varga **Theorem 2.3** (which follows immediately after) is de Bruijn (1950) [3] Theorem 1. It states that if $K_h(t) := e^{-h(t)}$ where h is even, $h(t) \ge 0$ for $t \in \mathbb{R}$, and $h'(t)$ is a uniform limit (on compact subsets of $\mathbb{C}$) of polynomials with zeros only on the imaginary axis, then $\int K_h(t) e^{izt}\, dt$ has only real zeros. This is closer to the log-concavity setting but requires the LP-class condition on $h'(t) = -(\log K)'(t)$, which is stronger than and not equivalent to $(\log K)'' \le 0$.

Neither Csordas–Varga Theorem 2.2 nor Theorem 2.3 directly states the strong log-concavity criterion of Theorem 1. Theorem 1 as stated has not been matched to a primary source.

The following primary sources have been examined or identified as candidates:

- **Pólya (1926), Acta Math. 48, 305–317** [8]. Proves that a *specific* modified Xi function $\Xi^*(z)$ has only real zeros by showing the entire function $\mathfrak{G}(iz; a) = \int_{-\infty}^{\infty} e^{-a(e^u + e^{-u}) + izu} du$ has only real zeros, using a difference equation and asymptotic analysis. Does *not* use log-concavity. English translation: Hosgood (2024), `https://translations.thosgood.com/AM-48-1926-305.html`.

- **Pólya (1926), J. London Math. Soc. 1, 98–99** [9]. A 2-page note, content not yet directly read. Cited by de Bruijn (1950) [3] as prior work on zeros of trigonometric integrals. *Candidate*: may contain an early version of the log-concavity criterion. Source task C-20 (open).

- **Newman (1976)** [6] proves $\Lambda_{DN} > -\infty$ (the de Bruijn–Newman constant is finite); not directly the log-concavity criterion.

$\square$

Remark 2 (Necessity of analyticity). Condition (v) is essential. The even function $K(t) = e^{-|t|^3}$ satisfies conditions (i)–(iv) but fails analyticity at the origin, and its cosine transform has 4 complex zeros in $[5, 15] \times [-5, 5]$ [2, Ex. 2.1]. Our computation independently confirms this (Attack 1 in Section 13). By contrast, e^{-t^p} for *even integer* p is real analytic and its cosine transform has only real zeros.

Remark 3 (Hypothesis verification for Theorem 1). The properties required to apply Theorem 1 to Φ are verified. For real t, evenness gives $F(t) = \int_{-\infty}^{\infty} \Phi(u)e^{itu}\,du = 2\int_0^{\infty} \Phi(u)\cos(tu)\,du = 2\,\Xi(t)$. Both $z \mapsto F(z)$ and $z \mapsto 2\Xi(z)$ are entire and agree on $\mathbb{R}$, so $F(z) = 2\Xi(z)$ for all $z \in \mathbb{C}$ by the identity theorem.

Cond. of Thm 1	Verified as	Where proved	Reference
(i) $K(t) > 0$	$\Phi(u) > 0$	Prop. 4(i)	[2, Thm. A]
(ii) $K \in L^1(\mathbb{R})$	$\Phi \in L^1(\mathbb{R})$	Prop. 4(iii)	[12, §2.10]
(iii) $(\log K)'' \le 0$	$(\log \Phi)'' < 0$ (strict)	Theorem 15	§§7–9
(iv) $O(e^{-\|t\|^{2+\delta}})$	$\Phi = O(e^{-\pi e^{2u}}) = O(e^{-u^3})$	Prop. 4(iv)	$u^3/e^{2u} \le 27/(8e^3) < \pi$
(v) Real analytic near 0	Φ real analytic on $\mathbb{R}$	Prop. 4(v)	Weierstrass M-test
(evenness) $K(-t) = K(t)$	$\Phi(-u) = \Phi(u)$	Prop. 4(ii)	[12, §2.10]

All six conditions of Theorem 1 are satisfied by Φ. See Appendix C for a hypothesis comparison table.

4 Properties of Φ

Proposition 4. *The kernel* Φ *defined by* (3) *satisfies:*

(i) $\Phi(u) > 0$ *for all* $u \ge 0$*;*

(ii) $\Phi(-u) = \Phi(u)$ *for all* u*;*

(iii) $\Phi \in L^1(\mathbb{R})$*;*

(iv) $\Phi(u) = O\big(e^{-\pi e^{2u}}\big)$ *as* $u \to +\infty$*;*

(v) Φ *is real analytic on* $\mathbb{R}$*.*

Proof. Properties (i)–(iii) are classical; see [12, §2.10] and [2, Theorem A]. Property (iv) follows from the dominant exponential factor $e^{-\pi e^{2u}}$ in φ_1. For (v): fix any $u_0 \in \mathbb{R}$. For $r = \pi/8$, any z with $|z - u_0| < r$ satisfies $\mathrm{Re}(e^{2z}) \ge e^{2(u_0 - r)}/\sqrt{2} =: c > 0$. Thus $|\varphi_n(z)| \le A\, n^4\, e^{-\pi c n^2/2}$ uniformly on $|z - u_0| \le r/2$. Since $\sum n^4 e^{-\pi c n^2/2} < \infty$, the Weierstrass M-test gives uniform convergence; hence Φ is real analytic at u_0. (Φ is *not* entire; only real analyticity on $\mathbb{R}$, as required by Condition (v), is claimed.)

Numerical verification. $\Phi(u) > 0$ at 10,001 uniformly spaced points on $[0, 1]$ (min $= 5.51 \times 10^{-7}$ at $u = 1$); $|\Phi(u) - \Phi(-u)|/|\Phi(u)| < 10^{-70}$ at 8 test points. □

5 Algebraic Core

Theorem 5 (Algebraic Core)**.** $(\log \varphi_1)''(u) < 0$ *for all* $u \ge 0$*.*

Proof. Write $\varphi_1 = g \cdot E$ where $g(u) = \pi e^{5u/2} h(u)$, $h(u) = 2\pi e^{2u} - 3$, and $E(u) = e^{-\pi e^{2u}}$. Since $h(u) \geq 2\pi - 3 > 0$ for $u \geq 0$, we have $\varphi_1 > 0$. Then

$$\log \varphi_1 = \log \pi + \tfrac{5}{2} u + \log h(u) - \pi e^{2u}, \quad (\log \varphi_1)'' = (\log h)'' - 4\pi e^{2u}.$$

With $h' = 4\pi e^{2u}$ and $h'' = 8\pi e^{2u}$:

$$(\log h)'' = \frac{h'' h - (h')^2}{h^2} = \frac{-24\pi e^{2u}}{h(u)^2} < 0.$$

Both terms are negative, so $(\log \varphi_1)'' < 0$. $\square$

Remark 6. $(\log \varphi_1)''(0) \approx -19.56$, monotonically decreasing. The series for $\Phi''(0)$ converges to $-68.052\ldots$ by $N = 5$ terms.

6 Tail Estimate

Lemma 7. *For $n \geq 2$ and $u \geq 0$: $e^{-\pi n^2 e^{2u}} \leq e^{-3\pi} \cdot e^{-\pi e^{2u}}$.*

Proof. Equivalent to $(n^2 - 1)e^{2u} \geq 3$, holding since $n^2 - 1 \geq 3$ and $e^{2u} \geq 1$. $\square$

Proposition 8. *For $u \geq 0$: $|R(u)|/\varphi_1(u) < 1/50$, where $R = \sum_{n \geq 2} \varphi_n$.*

Proof. $|\varphi_n|/\varphi_1 \leq 2n^4 e^{-\pi(n^2-1)e^{2u}}$ (since $B_n(u)/n^4 < 2$). At $u = 0$: $\sum_{n \geq 2} 2n^4 e^{-\pi(n^2-1)} < 0.006 < 1/50$. $\square$

Remark 9. The interval arithmetic retains $n = 1, \ldots, 5$ and omits $\delta := 4 \sum_{n \geq 6} \varphi_n$. Truncation errors on $[0, 1]$: $|\delta| \leq 7.03 \times 10^{-43}$, $|\delta'| \leq 1.18 \times 10^{-39}$, $|\delta''| \leq 1.98 \times 10^{-36}$, certified by interval arithmetic. At $u = 1$, the dominant cross term $|\delta'' \cdot \Phi_5| \approx 1.09 \times 10^{-42}$ gives total error $\leq 1.15 \times 10^{-42}$. Since $|Q_{\Phi_5}| \geq 3.36 \times 10^{-12}$, the safety factor exceeds 2.9×10^{30}.

7 Interval Arithmetic Verification

Definition 10. The *log-concavity numerator* of a positive function f is $Q_f(u) := f''(u) f(u) - f'(u)^2$.

Theorem 11. *$Q_\Phi(u) < 0$ for all $u \in [0, 1]$.*

Proof. Closed-form derivatives:

$$\begin{aligned} g_n' &= 9\pi^2 n^4 e^{9u/2} - \tfrac{15}{2} \pi n^2 e^{5u/2}, & g_n'' &= \tfrac{81}{2} \pi^2 n^4 e^{9u/2} - \tfrac{75}{4} \pi n^2 e^{5u/2}, \\ E_n' &= -2\pi n^2 e^{2u} \cdot E_n, & E_n'' &= \left(-4\pi n^2 e^{2u} + 4\pi^2 n^4 e^{4u}\right) \cdot E_n. \end{aligned}$$

All computations use `mpmath.iv` at 60-digit precision, retaining $n = 1, \ldots, 5$. Partition $[0, 0.949]$ into 1,898 subintervals and $[0.949, 1.0]$ into 51,000; on each, evaluate Q_{Φ_5} via interval arithmetic and certify the upper bound is negative.

Result: All 52,898 subintervals certified $Q_{\Phi_5}(u) < 0$, maximum upper bound -3.36×10^{-12}. Since truncation error $\leq 1.15 \times 10^{-42}$ (Remark 9), $Q_\Phi(u) < 0$ on $[0, 1]$. Cross-validation at 80-digit precision at 10 selected points: all within enclosures. $\square$

8 Extended Certification on $[1.0, 3.0]$

Direct interval arithmetic on Q_Φ fails for $u > 1$ due to catastrophic cancellation ($40\times$). We certify $(\log\Phi)''(u) < 0$ directly.

Lemma 12 (Uniform tail bound)**.** *For all $u \geq 1$: $|W_{\text{tail}}(u)| \leq G(1) = 1.82 \times 10^{-25}$, where $G(u) := \lambda(u) \cdot |W_1(u)|$ and $\lambda(u) := |\Delta Q(u)|/|Q_{\varphi_1}(u)|$.*

Proof. We show $G'(u) < 0$ for $u \geq 0$, so $G(u) \leq G(1)$ for $u \geq 1$.

$$\frac{d}{du}\log\lambda \leq 2 - 6\pi e^{2u} \quad \text{(dominant } n = 2 \text{ term decays as } e^{-6\pi e^{2u}}\text{)},$$
$$\frac{d}{du}\log|W_1| \leq 2 \quad (\text{since } |W_1| \geq 4\pi e^{2u} \text{ and } \frac{d}{du}|W_1| \leq 8\pi e^{2u}).$$

Hence $\frac{d}{du}\log G \leq 4 - 6\pi e^{2u} \leq 4 - 6\pi < 0$ for all $u \geq 0$. Therefore $G(u) \leq G(1) = 1.95 \times 10^{-27} \times 93.15 \approx 1.82 \times 10^{-25}$. □

Theorem 13. *$(\log\Phi)''(u) < 0$ for all $u \in [1.0, 3.0]$.*

Proof. Decompose $(\log\Phi)'' = W_1 + W_{\text{tail}}$ where $W_1(u) = -24\pi e^{2u}/h(u)^2 - 4\pi e^{2u}$ is proved negative in Theorem 5, and $|W_{\text{tail}}| \leq C\varepsilon^*$ with $C = 204$ and $\varepsilon^*(u) := 2\sum_{n=2}^{\infty} n^4\, e^{-\pi(n^2-1)e^{2u}}$. The constant is explicitly computed:

$$C := \left.\frac{|\Delta Q|}{\varepsilon \cdot |Q_{\varphi_1}|}\right|_{u=1} = 204, \quad \varepsilon(1) = 9.59 \times 10^{-30}, \quad |\Delta Q|/|Q_{\varphi_1}| = 1.95 \times 10^{-27}.$$

For each $i = 0, \ldots, 100$, interval enclosures of W_1 and ε over $I_i = [u_i - 0.01,\, u_i + 0.01]$ (60-digit precision) certify $\text{upper}(W_1(I_i)) + 204 \cdot \text{upper}(\varepsilon(I_i)) < 0$. Adjacent intervals overlap; their union $[0.99, 3.01] \supset [1.0, 3.0]$.

Result: All 101 interval checks certified, minimum margin 91.3 (at $u = 1.0$). By Lemma 12, $|W_{\text{tail}}| \leq 1.82\times10^{-25}$ on $[1, 3]$. Hence $(\log\Phi)'' \leq -93.1 + 1.82\times10^{-25} < 0$. Script: `proof/verify_ia_1_to_3.py` (SHA256: `1BB9E9DECF13580C...`). □

9 Tail Bound for $u \geq 3.0$

Theorem 14. *$(\log\Phi)''(u) < 0$ for all $u \geq 3.0$.*

Proof. At $u = 3$: $W_1(3) < -4\pi e^6 < -1000$ and $\varepsilon^*(3) \leq 32\, e^{-3\pi e^6} < 10^{-1636}$, so $W_1(3) + 204\,\varepsilon^*(3) < 0$.

For $u > 3$: $W_1'(u) < 0$ holds since $h^3 - 6h - 36 > 0$ for $h \geq 3.92$ (a fortiori for $u \geq 3$); each term of ε^* is strictly decreasing term-by-term. Since both W_1 and $204\varepsilon^*$ are strictly decreasing, $W_1(u) + 204\,\varepsilon^*(u) \leq W_1(3) + 204\,\varepsilon^*(3) < 0$.

Table: $\varepsilon^*(1.0) \approx 1.82 \times 10^{-29}$, $\varepsilon^*(1.5) \approx 1.96 \times 10^{-81}$, $\varepsilon^*(2.0) \approx 1.07 \times 10^{-222}$, $\varepsilon^*(3.0) < 10^{-1636}$. □

10 Main Result

Theorem 15 (Log-concavity of the Riemann–Jacobi kernel)**.** *The kernel Φ defined by* (3) *satisfies $(\log\Phi)''(u) < 0$ for all $u \geq 0$.*

Proof. Combine Theorems 11 ($u \in [0, 1]$), 13 ($u \in [1.0, 3.0]$), and 14 ($u \geq 3.0$). $\square$

Remark 16 (Conditional consequence under Pólya-type criterion). **Status: conditional on Verdict D (Section 2).**

If Theorem 1 is accepted as an unconditional result (i.e. a primary source with matching hypotheses is identified and verified), then the six properties of Theorem 1 verified for Φ in Remark 3 yield: the Fourier transform $F(z) = 2\Xi(z)$ has only real zeros, so Ξ has only real zeros, which is equivalent to RH [12].

This consequence is contingent on the source dependency catalogued in Section 12. **No unconditional Riemann Hypothesis claim is made.**

11 Conditional Branches

The RH implication depends on which form of the Pólya-type criterion holds.

Branch A (Verdict A hypothetical): Strong log-concavity criterion. Assume: if K is even, positive, integrable, real analytic near the origin, superexponentially decaying, and log-concave on $[0, \infty)$, then its Fourier transform has only real zeros. Under this assumption, properties (1)–(6) verified for Φ (Remark 3) yield RH. *This branch is conditional on locating a primary source supporting this criterion. Current source audit: Verdict D.*

Branch B (Satz I reading): Preservation theorem. Under the Satz I reading, multiplying by an $\mathcal{LP}$-factor preserves real-rootedness, but cannot establish it from scratch. Real-rootedness of Ξ cannot be derived without a prior real-rooted starting point. *Branch B does not yield RH from log-concavity of* Φ.

Branch C (Satz II reading): Mellin–Fourier theorem. Under Satz II, the hypothesis is that $H(z) = \int_0^\infty u^{z-1}\Phi(u)\,du$ has only real negative zeros, which is a separate non-trivial analytic condition not established here. *Branch C does not yield RH from log-concavity of* Φ *without additional work.*

Current status. Source audit (Section 2): Verdict D. Unconditional contribution: Theorem 15. The RH implication is an open conditional consequence pending source resolution.

12 Proof-Critical Dependencies

Claim	Type	Primary source	Status
$\varphi_1(u) > 0$ for $u \geq 0$	Lean	`phi1_pos`	PROVED
$(\log \varphi_1)''(u) < 0$ for $u \geq 0$	Lean	`log_phi1_d2_neg`	PROVED
$\Phi(u) > 0$ for $u \geq 0$	Cited	[2, Thm. A]	CITED
$\Phi(-u) = \Phi(u)$	Cited	[12, §2.10]	CITED
$\Phi \in L^1(\mathbb{R})$	Cited	[12, §2.10]	CITED
Φ real analytic on $\mathbb{R}$	Structural	Weierstrass M-test	PROVED
$\Phi = O(e^{-\pi e^{2u}})$	Structural	`phi1_decay_bound`	PROVED
$Q_\Phi(u) < 0$ on $[0, 1]$	IA cert	mpmath.iv + Arb/FLINT	CERTIFIED
$(\log \Phi)''(u) < 0$ on $[1.0, 3.0]$	Alg.+pert.	101-checkpoint script	CERTIFIED
$(\log \Phi)''(u) < 0$ for $u \geq 3$	Monotonicity	$\varepsilon^*(3) < 10^{-1636}$	PROVED
Pólya-type criterion (Thm 1)	Cited	[2] Thm 2.3 / [3]	CITED (Verdict D)
RH $\Leftrightarrow$ Ξ real zeros	Cited	[12, §2.1]	CITED

Status key: PROVED = formal (Lean) or elementary argument; CERTIFIED = rigorous computational certificate; CITED = external theorem with explicit hypothesis verification; CITED (Verdict D) = cited with source audit finding hypothesis mismatch.

The unconditional items (PROVED and CERTIFIED) establish Theorem 15. The conditional RH consequence additionally requires the CITED (Verdict D) item.

13 Falsification Testing

We subjected every proof-chain link to 36 systematic tests in 7 batches. No test detected any inconsistency.

#	Target	Attack	Result
1	Analyticity	e^{-t^3} cosine transform complex zeros	4 found
2	$\Phi > 0$	Search 10,001 points on $[0, 1]$	min $= 5.5 \times 10^{-7}$
4	$Q_\Phi < 0$	7,001 points (dense + random + adversarial)	All negative
6	Convention	$\int \Phi \, du = \xi(1/2)$	Ratio = 1.000
7	Constant C	Compute C at $u = 1$	$C = 204$
10	IA correctness	Containment: π, e, $e^{-\pi}$, compound	All pass
22	Product rule	Full φ_1'' vs. `mpmath.diff`	$< 10^{-15}$
24	$\Xi(5)$	$\int \Phi \cos(5u) \, du$ vs. $\xi(\frac{1}{2} + 5i)$	Ratio = 1.000
27	Pólya on $e^{-\cosh t}$	Argument-principle count	Only real
31	Adversarial Q	15,001 points on $[0.5, 2.0]$	All negative

Attack 12 historically detected a real bug (wrong g'' coefficient). The remaining 24 attacks cover: Φ evenness, decay, smoothness, circularity, series convergence, end-to-end checks at γ_2, and extended attacks 33–36 (10^5-point dense Q_Φ scan; Pólya conditions (i)–(v); Jensen polynomial hyperbolicity; $\Lambda = 0$ consistency).

14 Reproducibility

Script	Purpose
`verify.py`	Full proof verification pipeline
`falsify.py`	36 falsification attacks (7 batches)
`proof/verify_logconcavity_rigorous.py`	IA certification (Theorem 11)
`proof/verify_algebraic_core.py`	Algebraic core and perturbation bound
`proof/verify_logconcavity_arb.py`	Independent Arb/FLINT reproduction
`lean4/RHProof/Basic.lean`	Lean 4 proof structure (0 `sorry`)

Lean 4 formalization. 13 project-specific axioms, 6 proved theorems, 0 `sorry`. Machine-checked against Mathlib commit `a6276f4c` (0 errors, 0 correctness warnings).

Lean status note. The Lean formalization verifies the algebraic dominant-term core and records remaining dependencies as explicit axioms. The correct statement is: "Lean verifies the algebraic core with zero `sorry` declarations; remaining dependencies are recorded as explicit axioms." The axiom `polya_theorem` encodes the Pólya-type criterion dependency.

Item	Lean proved?	Lean axiom?	External citation?	Computational cert?
$\varphi_1(u) > 0$	Yes	No	No	No
$(\log \varphi_1)'' < 0$	Yes	No	No	No
$\Phi > 0$ on $[0, 1]$	No	Axiom	[2]	IA cert
$Q_\Phi < 0$ on $[0, 1]$	No	Axiom	No	IA cert
$(\log \Phi)'' < 0$ on $[1, 3]$	No	Axiom	No	101-checkpoint
Pólya-type criterion	No	Axiom	[2]	No
RH corollary	No	Conditional only	No	No

15 Discussion

Strengths. The algebraic core is machine-checked in Lean 4. The interval arithmetic uses exact symbolic derivatives verified by two independent libraries (`mpmath.iv` and Arb/FLINT, 55,892 subintervals, 200-bit precision). The extended certification avoids catastrophic cancellation by working with $(\log \Phi)''$ directly. The constant $C = 204$ is computed explicitly; the tail bound margin exceeds 10^{1634} at $u = 3$.

Relation to prior claims. A preprint by Gershon (April 2026) establishes the same log-concavity result but contains a presentation error in the perturbation bound. Our treatment computes C explicitly and extends the certification to $[0, 3.0]$.

Gershon's preprint also incorrectly claims that $K(t) = e^{-t^4}$ is a counterexample to log-concavity implying real zeros of the Fourier transform. This is wrong: Csordas–Varga [2, Ex. 2.1] explicitly state that $\int e^{-t^p} \cos(zt)\, dt$ has only real zeros for every *even integer* p, a fact our computation independently confirms (Attack 27, e^{-t^4}: zero complex zeros in $[0, 20] \times [-3, 3]$ by argument-principle). The correct sharp counterexample is $K(t) = e^{-|t|^3}$, which satisfies conditions (i)–(iv) of Theorem 1 but fails the real-analyticity condition (v) ($|t|^3$ is not differentiable at the origin), and whose cosine transform has infinitely many non-real zeros.

Contextual implications are in Appendix A.

16 Limitations and Open Dependencies

1. Pólya-type criterion: Verdict D (mismatch). Theorem 1 is cited from Csordas–Varga [2, Thm. 2.2]. Direct reading of Pólya (1927) Satz I and Satz II (Section 2) reveals that neither theorem states the strong log-concavity criterion: Satz I is a preservation theorem; Satz II is a Mellin–Fourier theorem (hypothesis = Mellin transform has only real negative zeros). **Source audit verdict: D — Mismatch.** Lean axiom `polya_theorem` records this dependency. *Required before journal submission: identify a primary source for the strong log-concavity criterion with matching hypotheses.*

2. Computational certificates axiomatized in Lean. Axioms `ia_verification_0_to_1` and `ia_verification_1_to_3` are not derived within the proof assistant.

3. Fourier representation and classical properties are cited. The identity $\Xi(t) = \int_0^\infty \Phi(u)\cos(tu)\,du$, evenness, and integrability are from Titchmarsh [12] §2.10.

4. No specialist peer review. Submission to a journal and specialist peer review remain outstanding.

References

[1] George Csordas, Timothy S. Norfolk, and Richard S. Varga. The Riemann hypothesis and the Turán inequalities. *Transactions of the American Mathematical Society*, 296(2):521–541, 1986.

[2] George Csordas and Richard S. Varga. Integral transforms and the Laguerre–Pólya class. *Complex Variables*, 12:211–230, 1989.

[3] N. G. de Bruijn. The roots of trigonometric integrals. *Duke Mathematical Journal*, 17:197–226, 1950.

[4] Dimitar K. Dimitrov and Fábio R. Lucas. Higher order Turán inequalities for the Riemann ξ-function. *Proceedings of the American Mathematical Society*, 139:1013–1022, 2011.

[5] Michael Griffin, Ken Ono, Larry Rolen, and Don Zagier. Jensen polynomials for the Riemann zeta function and other sequences. *Proceedings of the National Academy of Sciences*, 116(23):11103–11110, 2019.

[6] Charles M. Newman. Fourier transforms with only real zeros. *Proceedings of the American Mathematical Society*, 61(2):245–251, 1976. Proves
$Lambda_{DN} > -$
$infty$: the de Bruijn–Newman constant is finite. Also proves a closure/survival result for real zeros under Gaussian perturbation. Foundation for the de Bruijn–Newman constant.

[7] Tristen Kyle Pierson. specsmith: Applied epistemic engineering toolkit for ai-assisted development, 2026.

[8] G. Pólya. Bemerkung über die Integraldarstellung der Riemannschen
xi-Funktion. *Acta Mathematica*, 48:305–317, 1926. English translation: T. Hosgood, `https://translations.thosgood.com/AM-48-1926-305.html`. Proves that a modified
xi^* function has only real zeros via difference equation methods for

$mathfrakG(z;a) =$
$inte^{-a(e^u+e^{-u})+zu}du$. Does NOT use log-concavity of the kernel.

[9] G. Pólya. On the zeros of certain trigonometric integrals. *Journal of the London Mathematical Society*, 1:98–99, 1926. A 2-page preliminary note. Not directly accessible without institutional access. Contextual evidence (Cardon 2001, Pref. library holdings at Trinity College Cambridge) suggests it is a brief announcement of the results fully developed in the Acta Math. 1926 paper (the
xi^* real-zeros result via
$mathfrakG(z;a)$), not a general log-concavity criterion. At 2 pages, it cannot contain a general theorem with proof. C-20 open: requires institutional library access to verify directly.

[10] G. Pólya. Über trigonometrische Integrale mit nur reellen Nullstellen. *Journal für die reine und angewandte Mathematik*, 158:6–18, 1927.

[11] Brad Rodgers and Terence Tao. The de Bruijn–Newman constant is non-negative. *Forum of Mathematics, Pi*, 8:e6, 2020.

[12] E. C. Titchmarsh. *The Theory of the Riemann Zeta-Function*. Oxford University Press, 2nd edition, 1986. Revised by D. R. Heath-Brown.

A Contextual Results

The following results are consequences of or evidence consistent with RH but *do not form part of the proof chain* in Section 12.

(i) Jensen polynomial hyperbolicity. [5] establish that $J_n^d(X)$ is hyperbolic for each fixed d. If Remark 16 were established unconditionally, this would extend to all $d, n \geq 0$ simultaneously. Numerically verified for $k = 1, \ldots, 8$ ($d = 2$) and $n = 1, \ldots, 7$ ($d = 3$).

(ii) De Bruijn–Newman constant. Conditional on the RH consequence, $\Lambda = 0$ (combining Rodgers–Tao [11] and Platt–Trudgian).

(iii) CvS spectral framework. Eigenvalue plateau at $N = 10$: $\log_{10}|\lambda_{\min}(c)|$ reaches -34.2 at $c = 53$. Included as independent numerical evidence only.

(iv) AEE assessment. The AEE framework [7] assigns 0.169/1.000 (non-critical failures: Lean certificates not derived within Lean, no peer review, Verdict D source mismatch).

B Proof Certificate Table

Step	Claim	Type	Status	Script / Source
C1	$\varphi_1 > 0$	Lean	PROVED	`Basic.lean: phi1_pos`
C2	$(\log \varphi_1)'' < 0$	Lean	PROVED	`Basic.lean: log_phi1_d2_neg`
C3	$\Phi > 0$, even, $\in L^1$	Classical	CITED	[12, 2]
C4	$Q_\Phi < 0$ on $[0, 1]$	IA (mpmath.iv)	CERTIFIED	`verify_logconcavity_rigorous.py`
C4b	C4 reproduced	IA (Arb/FLINT)	CERTIFIED	`verify_logconcavity_arb.py`
C5	$(\log \Phi)'' < 0$ on $[1, 3]$	Alg.+pert.	CERTIFIED	`verify_ia_1_to_3.py`
C6	$(\log \Phi)'' < 0$ for $u \geq 3$	Monotonicity	PROVED	`verify_algebraic_core.py`
C7	Φ real analytic	Structural	PROVED	Weierstrass M-test
C8	$\Phi = O(e^{-\pi e^{2u}})$	Structural	PROVED	`phi1_decay_bound`
C9	Pólya-type criterion	Cited (Verdict D)	CITED	[2] Thm 2.3 / [3]
C10	RH $\Leftrightarrow$ Ξ real zeros	Cited	CITED	[12, §2.1]

SHA256 hashes:

Cert	Output file	SHA256 prefix
C4	`results/verify_logconcavity_rigorous.json`	`0D0841DAB32396D9...`
C4b	`results/verify_logconcavity_arb.json`	`974B67CC58B96117...`
C5	`results/verify_ia_1_to_3.json`	`1BB9E9DECF13580C...`
Main	`verification/proof_certificate_v2.json`	`8B538345D589638A...`

Full hashes in `docs/certificate_hash_table.md`.

C Pólya Source Analysis

The source audit is in Section 2. For reference, the hypothesis comparison between Pólya (1927) Satz II and Csordas–Varga (1989) Theorem 2.2:

Pólya 1927 Satz II	Csordas–Varga 1989 Thm 2.2	Our Φ satisfies
f for $t > 0$, real-valued, abs. integrable	Even $K > 0$, in L^1	Props (i)–(iii)
$\lvert f(t)\rvert < Be^{-t^{1+\beta}}$	$K = O(e^{-\lvert t\rvert^{2+\delta}})$, $\delta > 0$	Prop. (iv)
f analytic near $t = 0$	K real analytic near origin	Prop. (v)
$H(z)$ **has only real negative zeros**	**Log-concavity** $(\log K)'' \leq 0$	Theorems 11,13,14
Conc: $\int f(t^{2q})e^{izt}dt$ has real zeros	Conc: $\int K(t)e^{izt}dt$ has real zeros	—

Extended Verdict D: Csordas–Varga Theorem 2.2 is also the Mellin–Fourier theorem. Direct reading of Csordas–Varga (1989) reveals that their Theorem 2.2 is a restatement of Pólya Satz II: the hypothesis is that the Mellin transform $H(z) = \int_0^\infty t^{z-1}K(t)\,dt$ has only real negative zeros, not log-concavity of K. Csordas–Varga cite this as "[23, p. 7]" where [23] is Pólya (1927). Their Theorem 2.3 is de Bruijn (1950) Theorem 1, which uses an LP-class condition on $h'(t) = -(\log K)'(t)$, not simply $(\log K)'' \leq 0$.

Therefore the citation chain is broken at *both* ends:

- Pólya (1927): Satz I and Satz II do not state the log-concavity criterion.

- Csordas–Varga (1989) Theorem 2.2: this is Pólya Satz II (Mellin–Fourier), not the log-concavity criterion.

The strong log-concavity criterion (positivity + log-concavity + analyticity + decay ⇒ real zeros of Fourier transform) has not been matched to any primary source so far examined.

D Acceptance Checklist

The following criteria must all be satisfied for journal submission.

C-1. **Theorems proved or cited with explicit hypothesis matching.** ✓ complete, with Pólya-type criterion carrying Verdict D.

C-2. **Computational claims carry rigorous certificates.** ✓ complete.

C-3. **Lean build: zero `sorry`, zero correctness warnings.** ✓ confirmed (`lake build`, Mathlib `a6276f4c`).

C-4. **Lean certificates derived within the proof assistant.** × NOT MET. Axioms `ia_verification_0_to_1` and `ia_verification_1_to_3` remain unformalized.

C-5. **Pólya-type criterion formalized in Lean.** × NOT MET. Axiom `polya_theorem` is not machine-checked.

C-6. **Specialist peer review.** × NOT MET.

C-7. **Analytic consequences conditioned on unverified dependencies.** ✓ complete. RH consequence is Remark 16 (conditional only). Appendix A labels all consequences conditional.

C-8. **No OPEN entries in Proof-Critical Dependencies.** ∼ PARTIAL. Pólya criterion carries CITED (Verdict D).

C-9. **Φ analyticity restricted to real analyticity.** ✓ complete (Proposition 4(v)).

C-10. **Uniform $|W_{\mathrm{tail}}|$ bound proved.** ✓ PROVED (Lemma 12).

C-11. **Interval coverage for Theorem 13 explicit.** ✓ union $[0.99, 3.01] \supset [1, 3]$.

C-12. **Monotonicity for Theorem 14 proved.** ✓ PROVED.

C-13. **$F(z) = 2\Xi(z)$ identity by analytic continuation.** ✓ complete (Remark 3).

C-14. **Abstract and introduction language conditional on Verdict D.** ✓ complete. No unconditional RH claim is made.

C-15. **Certificate SHA256 hashes in Appendix B.** ✓ complete.

C-16. **Pólya Satz II source audit with verdict. ✓ COMPLETE — Verdict D (Mismatch).** Scanned original (GDZ LOG_0006) directly read. Satz I (p. 7): preservation theorem. Satz II (pp. 7–8): Mellin–Fourier theorem; hypothesis = Mellin transform $H(z)$ has only real negative zeros; log-concavity of f does not appear. The strong log-concavity criterion does not appear in Pólya (1927). Exact German text transcribed in Section 2. RH claim downgraded to conditional (Remark 16).

C-17. **Tail prefactor corrected from n^4 to $2n^4$.** ✓ RESOLVED.

C-18. **Script renamed to `verify_ia_1_to_3.py`.** ✓ DONE.

C-19. **"Do not claim" checklist.** ✓ The paper does not claim: "RH is proved"; "Pólya 1927 directly implies the criterion as stated" (Verdict D); "Lean verifies the proof of RH" (correct: Lean verifies the algebraic core); "No open dependencies remain"; "The Pólya source audit is complete" (it is; finding is Verdict D); "Computational certificates are formalized in Lean".

C-20. **Primary source for strong log-concavity criterion identified.** × NOT MET. Required: locate a peer-reviewed primary source stating the criterion (positivity + log-concavity + analyticity + decay ⇒ real zeros of Fourier transform) with hypotheses matching those verified for Φ. Candidates examined:

- Pólya (1926) Acta Math. 48 (read): proves Ξ^* has real zeros via difference equations for $\mathfrak{G}$; no log-concavity.
- Pólya (1927) J. reine angew. Math. 158 (read directly): Satz I = preservation; Satz II = Mellin–Fourier; no log-concavity.
- Csordas–Varga (1989) Thm 2.2 (read directly): Satz II restatement; Thm 2.3 = de Bruijn Thm 1 with LP-class condition on h'.
- Pólya (1926) J. London Math. Soc. 1 [9] (not directly accessible; paywalled; requires institutional library access): 2-page note. Contextual evidence from Cardon (2001) and Trinity College Cambridge library holdings suggests it announces the same ξ^* result as the Acta Math. 1926 paper (real zeros via $\mathfrak{G}(z;a)$ difference equations), not a general log-concavity criterion. A 2-page note cannot contain a general theorem with proof of the required scope. **Priority: obtain via institutional access.**
- de Bruijn (1950) [3]: read via Csordas–Varga; uses LP-class condition on h', stronger than log-concavity.

Overall: 14 of 20 criteria met. Open: C-4 (Lean IA), C-5 (Lean Pólya), C-6 (peer review), C-8 (partial), C-20 (primary source for log-concavity criterion). C-16 complete: Verdict D from direct German source reading.